

ICS MATH PART 1 - CHAPTER 1: NUMBER SYSTEMS

Punjab Textbook Board | Zehniyaat PK Free Notes

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SECTION A: IMPORTANT DEFINITIONS

1. Natural Numbers (N): $\{1, 2, 3, 4, \dots\}$

2. Whole Numbers (W): $\{0, 1, 2, 3, \dots\}$

3. Integers (Z): $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

4. Rational Numbers (Q): Numbers that can be written as p/q where $p, q \in \mathbb{Z}$ and $q \neq 0$

Examples: $1/2, -3/4, 5, 0.75, 0.333\dots$

5. Irrational Numbers: Non-terminating, non-repeating decimals

Examples: $\sqrt{2}, \sqrt{3}, \pi, e$

6. Real Numbers (R): Rational + Irrational numbers

7. Complex Numbers (C): $a + ib$ where $i = \sqrt{-1}$

SECTION B: EXERCISE 1.1 SOLVED

Q1. Identify which are rational and which are irrational:

(i) $\sqrt{4} = 2 \rightarrow$ Rational

(ii) $\sqrt{7} \rightarrow$ Irrational

(iii) $3.14 \rightarrow$ Rational

(iv) $\pi \rightarrow$ Irrational

(v) $0 \rightarrow$ Rational

(vi) $22/7 \rightarrow$ Rational

Q2. Express in p/q form:

(i) $0.6 = 6/10 = 3/5$

(ii) $0.777\dots$

Let $x = 0.777\dots$

$10x = 7.777\dots$

$9x = 7 \rightarrow x = 7/9$

(iii) $0.232323\dots$

Let $x = 0.2323\dots$

$100x = 23.2323\dots$

$99x = 23 \rightarrow x = 23/99$

Q3. Write 3 rational numbers between 3 and 4:

Ans: $3.1 = 31/10, 3.5 = 7/2, 3.75 = 15/4$

SECTION C: EXERCISE 1.2 - LAWS OF EXPONENTS

Important Laws:

1. $a^m \times a^n = a^{(m+n)}$

2. $a^m \div a^n = a^{(m-n)}$

3. $(a^m)^n = a^{mn}$

4. $a^0 = 1$, where $a \neq 0$

5. $a^{-m} = 1/a^m$

Q1. Simplify: $2^3 \times 2^4$

Solution: $2^{(3+4)} = 2^7 = 128$

Q2. Simplify: $(3^2)^3$

Solution: $3^{(2 \times 3)} = 3^6 = 729$

Q3. Simplify: $5^7 \div 5^4$

Solution: $5^{(7-4)} = 5^3 = 125$

SECTION D: EXERCISE 1.3 - COMPLEX NUMBERS

$i = \sqrt{-1}$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$

Q1. Simplify: $(2 + 3i) + (4 - i)$

Solution: $(2+4) + (3i-i) = 6 + 2i$

Q2. Simplify: $(3 + 2i)(1 - i)$

Solution: $3(1) + 3(-i) + 2i(1) + 2i(-i)$

$= 3 - 3i + 2i - 2i^2$

$= 3 - i - 2(-1) = 3 - i + 2 = 5 - i$

Q3. Find conjugate of $4 - 3i$

Solution: $4 + 3i$

SECTION E: IMPORTANT FORMULAS SHEET

1. $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

2. $\sqrt{a} / \sqrt{b} = \sqrt{(a/b)}$

3. $(a + b)^2 = a^2 + 2ab + b^2$

4. $(a - b)^2 = a^2 - 2ab + b^2$

5. $a^2 - b^2 = (a+b)(a-b)$

6. $\sqrt{50} = \sqrt{(25 \times 2)} = 5\sqrt{2}$

7. $\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$

SECTION F: PAST PAPER QUESTIONS 2019-2024

Q1. [Lahore Board 2023] Simplify: $\sqrt{18} + \sqrt{50}$

Solution: $\sqrt{18} + \sqrt{50} = \sqrt{(9 \times 2)} + \sqrt{(25 \times 2)} = 3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$

Q2. [Multan Board 2022] Write in standard form: $(2 + i)(3 - 2i)$

Solution: $6 - 4i + 3i - 2i^2 = 6 - i - 2(-1) = 6 - i + 2 = 8 - i$

Q3. [Faisalabad Board 2024] If $x = 2 + \sqrt{3}$, find $x + 1/x$

Solution: $1/x = 1/(2+\sqrt{3}) \times (2-\sqrt{3})/(2-\sqrt{3}) = (2-\sqrt{3})/(4-3) = 2-\sqrt{3}$

$x + 1/x = (2+\sqrt{3}) + (2-\sqrt{3}) = 4$

Q4. [Gujranwala Board 2021] Express 0.585858... in p/q form

Solution: Let $x = 0.5858\dots$

$$100x = 58.5858\dots$$

$$99x = 58 \rightarrow x = 58/99$$

SECTION G: MCQs FOR QUICK REVISION

1. π is a/an ____ number.

- a) Rational b) Irrational c) Integer d) Natural

Ans: b) Irrational

2. Value of i^4 is:

- a) 1 b) -1 c) i d) -i

Ans: a) 1

3. Conjugate of $3 + 4i$ is:

- a) $3 - 4i$ b) $-3 + 4i$ c) $-3 - 4i$ d) $4 + 3i$

Ans: a) $3 - 4i$

4. $2^0 = ?$

- a) 0 b) 1 c) 2 d) Undefined

Ans: b) 1

5. $\sqrt{16} + \sqrt{9} = ?$

- a) 5 b) 7 c) 25 d) $\sqrt{25}$

Ans: b) 7

6. Which is not rational?

- a) 0.5 b) $\sqrt{4}$ c) $\sqrt{5}$ d) $3/7$

Ans: c) $\sqrt{5}$

7. $(a^m)^n = ?$

- a) $a^{(m+n)}$ b) a^{mn} c) $a^{(m-n)}$ d) $a^{(m/n)}$

Ans: b) a^{mn}

8. If $z = a + ib$, then $|z| = ?$

- a) $a + b$ b) $\sqrt{a^2 + b^2}$ c) $a^2 + b^2$ d) $a - b$

Ans: b) $\sqrt{a^2 + b^2}$

9. $3\sqrt{2} + 5\sqrt{2} = ?$

- a) $8\sqrt{2}$ b) $15\sqrt{2}$ c) $8\sqrt{4}$ d) $15\sqrt{4}$

Ans: a) $8\sqrt{2}$

10. $1/(2+\sqrt{3}) = ?$

- a) $2+\sqrt{3}$ b) $2-\sqrt{3}$ c) $\sqrt{3}-2$ d) $-2-\sqrt{3}$

Ans: b) $2-\sqrt{3}$

SECTION H: EXAM TIPS

1. Rationalize denominator hamesha - multiply by conjugate
2. Complex numbers mein $i^2 = -1$ use karna na bhoolo
3. MCQs mein π , e , $\sqrt{2}$, $\sqrt{3}$ hamesha irrational hote hain
4. 0 rational number hai kyun ke $0/1$ likh sakte hain
5. Square root ko simplify karo: $\sqrt{72} = 6\sqrt{2}$

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